DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

UNIVERSITY COLLEGE OF ENGINEERING (A)

Osmania University, Hyderabad-07

CLASS TEST-I

SUBJECT:BASIC ELECTRONICS

Class: B.E. (CSE)

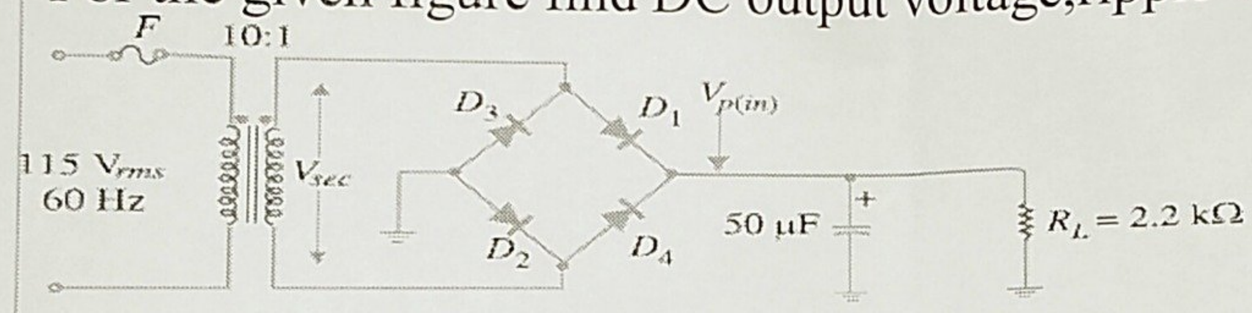
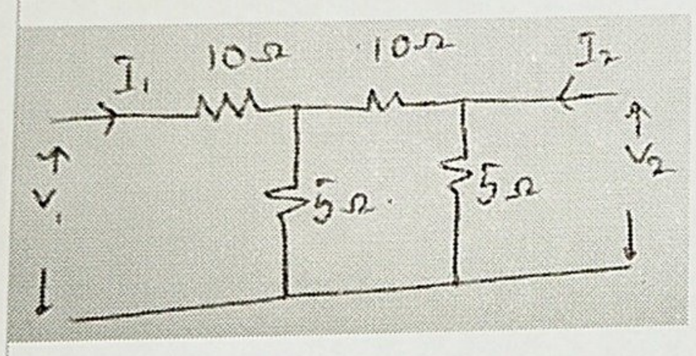
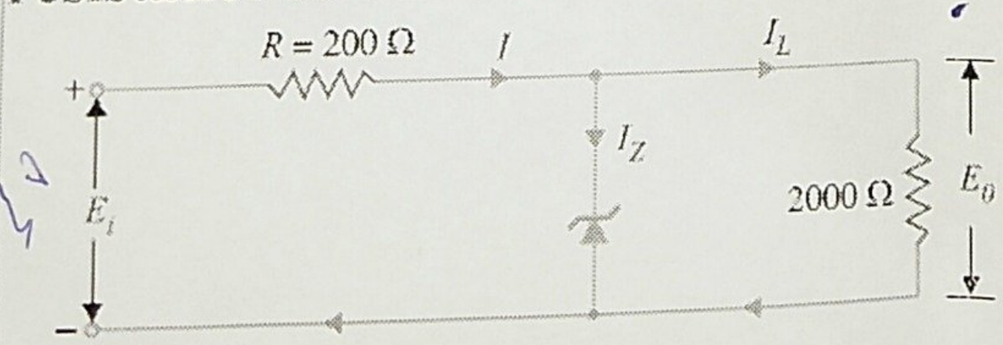
Semester: III

Academic Year: 2025-26

Max Marks: 30

Duration:1 hr 30 min

Date:24-09-2025

		Marks	BT	CO
	Answer All Questions PART A (5 x 2 = 10 M)			
1. a)	What do you mean by ripple factor ? Give its significance	2	2	2
b)	Differentiate avalanche and zener breakdown	2	2	2
c)	What is the purpose of coupling capacitors ?	2	2	1
d)	Give the block diagram of DC power supply ?	2	1	2
e)	Draw the symbols of npn and pnp transistor and give the signifance of arrow? .	2	2	2
	Answer any two PART B (2 x 10 = 20 M)			
2. a)	Derive the ripple factor for a centre tapped rectifier with L filter with relevant wave forms	5	2	2
b)	For the given figure find DC output voltage,ripple factor 	5	3	2
3. a)	Explain the working of transistor npn transistor with components	4	2	1
b)	Derive the h parameters of the given circuit. 	6	4	1
4.a)	**Over what range of input voltage will the zener circuit shown in Fig maintain 30 V across 2000 Ω load, assuming that series resistance R = 200 Ω and zener current rating is 25 mA ?** 	4	3	1
b)	**Explain the purpose of biasing a BJT and working of Self bias circuit with stability factor ?**	6	2	2

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